

Week 10 Revision Selected Questions

Worked Solutions

Based on Sem 2 Week 10 - Revision.pdf

1. GreenGrid Energy

Let $x_A, \dots, x_F \in \{0, 1\}$ where $x_i = 1$ if project i is selected.

(a) A suitable binary model is

$$\max Z = 14x_A + 17x_B + 11x_C + 20x_D + 9x_E + 15x_F$$

subject to

$$6x_A + 8x_B + 5x_C + 9x_D + 4x_E + 7x_F \leq 21$$

$$x_D \leq x_B$$

$$x_A + x_F \leq 1$$

$$x_C + x_E \geq 1$$

$$x_i \in \{0, 1\} \quad \text{for all projects } i.$$

(b) The optimal solution is

$$x_B = x_D = x_E = 1, \quad x_A = x_C = x_F = 0.$$

Total cost:

$$8 + 9 + 4 = 21$$

Total benefit:

$$17 + 20 + 9 = 46.$$

So the best project set is $\{B, D, E\}$ with maximum total annual benefit score 46.

(c) Binary variables are required because each project is indivisible. It is meaningful to choose or reject a project, but not to choose, for example, 0.4 of project D .

(d) Add the constraint

$$x_A + x_B + x_C + x_D + x_E + x_F = 4.$$

Under the budget and logic rules, there is no feasible solution selecting exactly four projects. So the revised model is infeasible.

(e) One realistic extension would be a multi-period capital budgeting model with separate yearly budgets and project timing decisions. This would need extra data on when costs are incurred, when benefits begin, and any project sequencing dependencies.

2. Hock-Brooke Automobiles

Because the revision sheet does not specify a random-number stream or a simulation horizon, more than one valid simulation answer is possible. Using the accompanying spreadsheet workbook, the intended approach is to show an illustrative short run and then estimate average cost from a much longer simulation. The short table below is retained as an example, and the long-run recommendation is aligned to the spreadsheet results.

- (a) For capacity 9, one valid 12-week simulation is:

Week	Opening Stock	Delivered	Demand	Sales	Closing Stock	Overflow	Cost
1	0	9	6	6	3	0	90
2	3	8	7	7	4	0	90
3	4	9	10	10	3	0	90
4	3	10	7	7	6	0	90
5	6	9	6	6	9	0	90
6	9	8	8	8	9	0	90
7	9	7	7	7	9	0	90
8	9	9	8	8	10	1	110
9	10	8	9	9	9	0	90
10	9	10	6	6	13	4	170
11	13	10	10	10	13	4	170
12	13	9	7	7	15	6	210

This gives a short-run 12-week cost of

$$1380$$

and a short-run average weekly cost of

$$\frac{1380}{12} = 115.$$

Using the spreadsheet logic over the long run, the estimated average weekly cost for capacity 9 is approximately

$$103 \text{ per week}$$

more precisely about 103.04 on the simulation sheet, and about 102.91 on the analysis sheet.

- (b) Using the spreadsheet analysis sheet, the estimated long-run average weekly costs are:

Capacity	Average Weekly Cost
6	110.74
7	104.09
8	102.16
9	102.91
10	106.89
11	114.32

So, based on the spreadsheet outputs, capacity 9 is not the best choice. The lowest long-run average cost is achieved at capacity 8, with an estimated average weekly cost of about 102.16. Capacity 9 is only slightly worse, but the spreadsheet evidence supports recommending 8 spaces rather than 9.

Note: because no specific random-number stream was given in the source sheet, other correctly executed simulation runs may give slightly different short-run estimates.

3. University Library Help Desk

- (a) Main components:

Entities: students.

Resource: advisers.

Events: arrival, service start, service completion.

Stochastic inputs: interarrival times and service times.

Performance measures: number served, average wait, maximum wait and the percentage waiting less than 8 minutes.

- (b) One valid baseline run gives:

Measure	Result
Students served	41
Average wait	1.17 minutes
Maximum wait	10.32 minutes
% waiting less than 8 minutes	95.12%

- (c) Peak-period results:

Advisers	Served	Avg wait	Max wait	% below 8 min
2	70	1.88	12.99	95.71%
3	73	0.32	3.22	100.00%
4	70	0.08	2.23	100.00%

The target is 97% waiting less than 8 minutes. The minimum staffing level that meets the target is therefore 3 advisers.

- (d) A **PriorityResource** can be used instead of an ordinary **Resource**. Each arrival is labelled priority or standard, for example with probability 0.10 of being priority. Priority users would request service with a smaller priority number, such as **priority=0**, while standard users might request with **priority=1**. This would generally reduce waits for priority users and increase waits for standard users.
- (e) A single run is not enough because simulation output is affected by random variation. Replications with different seeds give more stable estimates and allow the calculation of confidence intervals. If the system were modelled as an ongoing operation rather than a fixed 360-minute session, a warm-up period could also be used to reduce start-up bias.

4. F.B. Knight Deliveries

Because the costs are stated per 10 items, it is convenient to model the variables in tens of items. Then warehouse supplies are 10, 10 and 10, while shop demands are 7, 8 and 15.

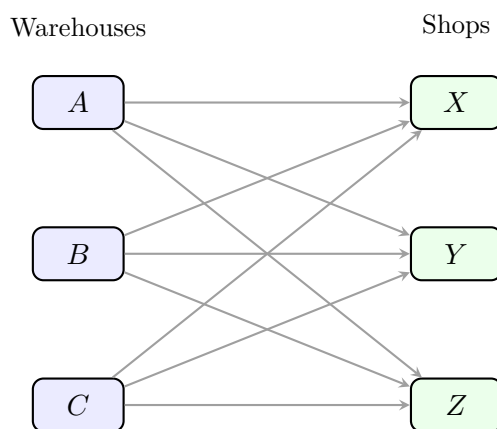
(a) Let

x_{ij} = number of tens of items shipped from warehouse i to shop j

for

$$i \in \{A, B, C\}, \quad j \in \{X, Y, Z\}.$$

The direct transportation network is:



The direct-shipping transportation model is

$$\min Z = 8x_{AX} + 7x_{AY} + 5x_{AZ} + 12x_{BX} + 3x_{BY} + 9x_{BZ} + 10x_{CX} + 11x_{CY} + 7x_{CZ}$$

subject to warehouse supply constraints

$$x_{AX} + x_{AY} + x_{AZ} = 10$$

$$x_{BX} + x_{BY} + x_{BZ} = 10$$

$$x_{CX} + x_{CY} + x_{CZ} = 10$$

shop demand constraints

$$x_{AX} + x_{BX} + x_{CX} = 7$$

$$x_{AY} + x_{BY} + x_{CY} = 8$$

$$x_{AZ} + x_{BZ} + x_{CZ} = 15$$

and non-negativity:

$$x_{ij} \geq 0.$$

One optimal direct-shipping plan is:

Route	Quantity (tens of items)	Cost per 10 items
$A \rightarrow Z$	10	5
$B \rightarrow X$	2	12
$B \rightarrow Y$	8	3
$C \rightarrow X$	5	10
$C \rightarrow Z$	5	7

Total cost:

$$10(5) + 2(12) + 8(3) + 5(10) + 5(7) = 183.$$

(b) With the restrictions

$$x_{BY} = 0$$

and

$$x_{AY} \geq 0.5$$

the same arc-variable model is modified by adding

$$x_{BY} = 0$$

and

$$x_{AY} \geq 0.5.$$

In tens-of-items units, one optimal revised plan is:

Route	Quantity (tens of items)
$A \rightarrow Y$	8
$A \rightarrow Z$	2
$B \rightarrow X$	7
$B \rightarrow Z$	3
$C \rightarrow Z$	10

Revised total cost:

$$8(7) + 2(5) + 7(12) + 3(9) + 10(7) = 247.$$

So the extra restrictions increase the cost from 183 to 247.

(c) If direct shipping is not allowed and the intermediary locations are used, define arc variables

y_{ik} = flow from warehouse i to intermediary k

for

$$i \in \{A, B, C\}, \quad k \in \{D_1, D_2\}$$

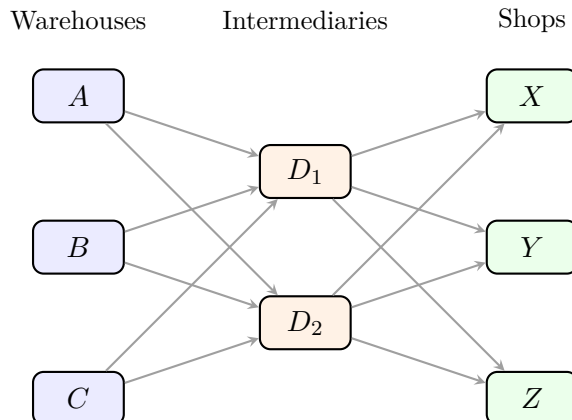
and

z_{kj} = flow from intermediary k to shop j

for

$$k \in \{D_1, D_2\}, \quad j \in \{X, Y, Z\}.$$

The transshipment network is:



The transshipment model is

$$\min Z = 4y_{AD_1} + 5y_{AD_2} + 3y_{BD_1} + 2y_{BD_2} + 6y_{CD_1} + 4y_{CD_2} + 4z_{D_1X} + 5z_{D_1Y} + 2z_{D_1Z} + 2z_{D_2X} + 4z_{D_2Y} + 4z_{D_2Z}$$

subject to supply constraints at the warehouses

$$y_{AD_1} + y_{AD_2} = 10$$

$$y_{BD_1} + y_{BD_2} = 10$$

$$y_{CD_1} + y_{CD_2} = 10$$

demand constraints at the shops

$$z_{D_1X} + z_{D_2X} = 7$$

$$z_{D_1Y} + z_{D_2Y} = 8$$

$$z_{D_1Z} + z_{D_2Z} = 15$$

and flow-balance constraints at the intermediary nodes

$$y_{AD_1} + y_{BD_1} + y_{CD_1} = z_{D_1X} + z_{D_1Y} + z_{D_1Z}$$

$$y_{AD_2} + y_{BD_2} + y_{CD_2} = z_{D_2X} + z_{D_2Y} + z_{D_2Z}.$$

All arc flows satisfy

$$y_{ik} \geq 0, \quad z_{kj} \geq 0.$$

If only one intermediary is used, then the variables on the unused node are omitted from the model. The three possibilities are:

Intermediary choice	Total cost
D_1 only	228
D_2 only	216
Both D_1 and D_2	181

So both intermediary locations should be used.

One optimal transshipment plan is:

Route	Quantity (tens of items)
$A \rightarrow D_1$	10
$B \rightarrow D_1$	5
$B \rightarrow D_2$	5
$C \rightarrow D_2$	10
$D_1 \rightarrow Z$	15
$D_2 \rightarrow X$	7
$D_2 \rightarrow Y$	8

Total cost:

$$10(4) + 5(3) + 5(2) + 10(4) + 15(2) + 7(2) + 8(4) = 181.$$

This is slightly cheaper than the best direct-shipping plan.

5. Forecasting at N&F

- (a) The quarterly data show a clear upward trend over time. Using centred 4-quarter moving averages, the smoothed values where available are:

Period	Centred 4-quarter moving average
3	1020.250
4	1030.500
5	1040.750
6	1050.625
7	1060.500
8	1070.375
9	1080.375
10	1090.625
11	1100.875
12	1111.125
13	1121.250
14	1131.000

- (b) Estimated seasonal variations:

Additive seasonal values:

$$\text{Spring} = 12.48, \quad \text{Summer} = 15.85, \quad \text{Autumn} = -18.27, \quad \text{Winter} = -10.06.$$

Multiplicative seasonal indices:

$$\text{Spring} = 1.0117, \quad \text{Summer} = 1.0147, \quad \text{Autumn} = 0.9829, \quad \text{Winter} = 0.9907.$$

- (c) A linear trend fitted to the centred moving averages is approximately

$$\hat{T}_t = 990.12 + 10.07t.$$

Forecasts for next year are therefore:

Period	Season	Additive Forecast	Multiplicative Forecast
17	Spring	1173.74	1174.83
18	Summer	1187.18	1188.53
19	Autumn	1163.12	1161.18
20	Winter	1181.40	1180.42

- (d) Both models give very similar forecasts. However, the additive model is slightly easier to justify because the seasonal effects appear roughly constant in size over time, rather than increasing proportionally with the level of the series. So an additive model would be a reasonable recommendation here, although either model would give very similar practical forecasts.

End of Solutions